

COMS 3003A

Test 2 – Deferred version

7 May, 2024

This is a closed book test.
Time allowance: 60 minutes

Caution: if, in any of your answers, you define a Turing machine and claim that it solves a particular problem, you have to prove your claim.

1. **(20 points)** Prove or disprove the following statements (i.e., if you believe the statement to be true, prove it; otherwise, provide a counterexample):

- (a) **(5 points)** If languages L_1 and L_2 are undecidable, then $L_1 \cup L_2$ is undecidable.

This is false. Let L_1 be any undecidable language. Then, $\bar{L}_1$ is also undecidable, but $L_1 \cup \bar{L}_1$ is decidable, since $L_1 \cup \bar{L}_1 = \Sigma^*$, and so it is decided by a Turing machine that accepts every input.

- (b) **(5 points)** Every countably infinite language is undecidable.

This is false. The countably infinite language Σ^* is clearly decidable: it is decided by a Turing machine that accepts every input.

- (c) **(5 points)** Every undecidable language is countably infinite.

This is true. Every language is countable. So, we need to show that every undecidable language is infinite. This is equivalent to every finite language being decidable. The latter is clearly true: if a language L is finite, it can be decided by a Turing machine that first writes out all the words in L and then checks if the input word is among them.

- (d) **(5 points)** If languages L_1 and L_2 are undecidable, then $L_1 \cap L_2$ is undecidable.

This is false. Let L_1 be any undecidable language. Then, $\bar{L}_1$ is also undecidable, but $L_1 \cap \bar{L}_1$ is decidable, since $L_1 \cap \bar{L}_1 = \emptyset$, and so it is decided by a Turing machine that rejects every input.

2. **(10 points)** Prove that the problem of determining whether the language of a Turing machine contains exactly 4 words is algorithmically unsolvable, i.e., that the language $\{\langle M \rangle : |L(M)| = 4\}$ is undecidable. You may assume that the following problems are unsolvable:

- non-self-acceptance for Turing machines;

- self-acceptance for Turing machines;
- acceptance for Turing machines.

Suppose, for the sake of contradiction, that this problem is solvable. Then, there exists a Turing machine, say M_4 , that solves it. Then, we can define a Turing machine M_A that decides acceptance:

- (1) M_A , on input $\langle M, w \rangle$, writes down (the program of) a Turing machine $M_{M,w}$ that accepts as input a binary word x and works as follows:
 - (a) $M_{M,w}$ simulates M on w ;
 - (b) If $M(w) = 1$, then $M_{M,w}$ accepts x if x is either 00 or 01 or 10 or 11; otherwise $M_{M,w}$ rejects x .
- (2) M_A calls M_4 with the argument $\langle M_{M,w} \rangle$.
- (3) If $M_4(\langle M_{M,w} \rangle) = 1$, then M_A accepts; $M_4(\langle M_{M,w} \rangle) = 0$, then M_A rejects.

Let's show that M_A , indeed, decides acceptance:

- If $M_A(\langle M, w \rangle) = 1$, then $M_4(\langle M_{M,w} \rangle) = 1$, which means, by specification of M_4 , that $|L(M_{M,w})| = 4$. This means that $L(M_{M,w}) \neq \emptyset$, which happens only if $M(w) = 1$.
- If $M_A(\langle M, w \rangle) = 0$, then $M_4(\langle M_{M,w} \rangle) = 0$, which means, by specification of M_4 , that $|L(M_{M,w})| \neq 4$. By definition of $M_{M,w}$, this happens only if $L(M_{M,w}) = \emptyset$, which means that $M(w) = 0$ or $M(w) = \infty$, i.e., that $M(w) \neq 1$.

Since acceptance is unsolvable, however, M_A does not exist, which gives us a contradiction.

3. (20 points) Does there exist a bijection between the set of all deciders and the set of all undecidable decision problems? Prove your statement. Your proof may not assume any fact proven in class or in textbook—if you need to use a fact in your proof, you need to prove it first.

No, such a bijection does not exist. To show this, we show that the set of deciders is countable, but the set of undecidable decision problems is uncountable.

Since every Turing machine, and hence every decider, can be encoded as a string in a finite alphabet, the set of deciders is countable.

We can show using diagonalization that the set of all decision problems is uncountable. The set of all decidable problems is countable, since the cardinality of this set cannot exceed the cardinality of the set of all deciders. If the set of undecidable problems was also countable, then we could combine two enumerations to obtain an enumeration of the set of all decision problems, which is impossible, since this set is uncountable.

4. (10 points) The first-order language of set theory contains a binary predicate symbol $\in$. Translate the following statements into this language (you are free to define any other symbols in terms of the ones listed above):

- (a) (5 points) There exists a set that contains exactly one element.

$$\exists x \exists y (y \in x \wedge \forall z (z \in x \rightarrow z = y)).$$

- (b) (5 points) For every pair of sets X and Y , there exists a set whose elements are the common elements of X and Y .

$$\forall x \forall y \exists z \forall u (u \in z \leftrightarrow u \in x \wedge u \in y).$$